Human-reviewed technical translation

junior_roller_skate_target_brief.pdf

Overall status	APPROVED WITH CORRECTIONS
Reviewer	Nina Doinjashvili
Reviewed	2 August 2026, 22:10 UTC

Approved as generated	Corrected	Needs further review	Validation errors
2	**4**	**0**	**0**

Reviewer decision

Approved after correcting the omitted winter-cracking intent, local citation coverage, recycled-content verification gap, and qualifier force in the trade-offs.

Compact artifact identity

`Source SHA-256: 43b394906f3f...b81f7300`

`Base report JSON SHA-256: 3a00227fc8da...8dc92fb0`

`Reviewed JSON SHA-256: e0c508fdb9fb...1cb37796`

Full hashes and the external PDF/Markdown manifest are referenced on the final page.

1. Product intent

Model-generated translation

Develop the rear brake housing for a junior roller skate used by children aged 6 to 12, with reliability as the first priority: it should provide at least the current protection and durability and must not feel less robust. Approximately 15% lower mass would be commercially valuable but is subordinate to reliability; the component should contain at least 30% recycled material by mass, with post-consumer content preferred, documented feedstock origin and traceability mandatory, and no significant durability decrease accepted as sustainability success. The visible surface should remain black and low-gloss without looking noticeably lower quality, while the team wants to avoid significant component-cost or manufacturing disruption; mould reuse is strongly preferred, and a cycle-time increase above 10% should be avoided.

Uncertainties

- “Significant” cost increase, manufacturing disruption and durability decrease, and “noticeably lower quality” are not defined in the brief.

Model critic assessment (before human review)

Revise: The translated claims preserve the cited priorities and qualifiers, but the product intent omits the explicitly identified main problem: winter-use cracking, particularly following impacts or falls. Although this appears in performance_to_validate, it is also central to why the product is being developed and should be retained compactly in product intent.

Semantic support: Direct

Issue types: Omission

Shared verified source evidence

"The company is developing the next generation of a rear brake housing for a junior roller skate used by children aged 6 to 12." Page 1
"The new component should provide at least the same level of protection and durability as the current version." Page 1
"Reliability is the first priority. The component must not feel less robust than the current one. A lighter part is valuable, and the visible surface must remain black and low-gloss." Page 2
"A mass reduction of approximately 15% would be commercially valuable, but reliability has higher priority." Page 1
"The component should contain at least 30% recycled material by mass." Page 1
"Post-consumer recycled content is preferred. Feedstock origin and traceability must be documented. Recycled content is not a success if durability decreases significantly." Page 2
"The visible surface should remain black and low-gloss and should not look noticeably lower quality." Page 1
"The team wants to avoid a significant increase in component cost or disruption to manufacturing." Page 1
"Existing mould Strong preference to reuse it Stakeholder preference" Page 1
"A cycle-time increase above 10% should be avoided." Page 2

Human review

CORRECTED

Human-reviewed translation

Develop the rear brake housing for a junior roller skate used by children aged 6 to 12, addressing the main problem of winter-use cracking, particularly after impacts or falls, with reliability as the first priority: it should provide at least the current protection and durability and must not feel less robust. Approximately 15% lower mass would be commercially valuable but is subordinate to reliability; the component should contain at least 30% recycled material by mass, with post-consumer content preferred, documented feedstock origin and traceability mandatory, and no significant durability decrease accepted as sustainability success. The visible surface should remain black and low-gloss without looking noticeably lower quality, while the team wants to avoid significant component-cost or manufacturing disruption; mould reuse is strongly preferred, and a cycle-time increase above 10% should be avoided.

Reviewed uncertainties

- “Significant” cost increase, manufacturing disruption and durability decrease, and “noticeably lower quality” are not defined in the brief.

Reason for correction

Restore the brief's central winter-cracking problem without changing the stated priority hierarchy.

Evidence added or removed

Added evidence

"The main problem to solve is cracking reported during winter use, particularly after impacts or falls."
Page 1

2. Component function

Model-generated translation

The housing protects and retains the roller skate’s rear braking mechanism.

Model critic assessment (before human review)

Supported: The statement directly translates the component-level function rather than substituting a material property. The omitted exposure conditions are service context, not additional component functions.

Semantic support: Direct

Translated wording: "The housing protects and retains the roller skate’s rear braking mechanism."

Shared verified source evidence

"The component protects and retains the rear braking mechanism and is exposed to drops, falls, kerbs and outdoor conditions."
Page 1

Human review

APPROVED AS GENERATED

No changes made. The model translation and shared evidence above are the reviewed result.

3. Performance to validate

Model-generated translation

Validation should demonstrate at least the current level of protection and durability across the approximately -10°C to 45°C outdoor-use range and assess cracking after relevant impacts or falls and deformation that could interfere with the brake mechanism; no numerical stiffness target exists. The need is supported by 17 prior-winter returns, including 13 from regions that had fallen below 0°C, with most cracks near the screw boss or wall-to-brake-support junction, and by an informal observation in which two of five housings cracked after complete rollers were conditioned for four hours at -10°C and dropped 1.0 metre onto concrete; actual field histories were unknown, and the observation lacked a room-temperature control, measured impact energy and a standardized protocol. Evaluation should also measure the proposed approximately 15% mass reduction against the confirmed 82 g current mass without displacing reliability, check the expected at-least-30% recycled content by mass, and assess whether the visible surface remains black and low-gloss without looking noticeably lower quality.

Assumptions

- The stated outdoor-use range and named impact or fall conditions are treated as relevant validation contexts, not as a prescribed test protocol.

Uncertainties

- The brief provides no numerical stiffness target, and the historical observations do not establish a standardized protocol or future cracking acceptance limit.

Model critic assessment (before human review)

Supported: The section states future-facing, observable product or component outcomes while clearly separating the field returns and informal cold-drop observation from future acceptance criteria. It preserves the approximate service range, comparative reliability requirement, absence of a stiffness target, historical counts, and limitations of the available observations without treating them as a validated protocol or cracking limit.

Semantic support: Direct

Translated wording: "Validation should demonstrate at least the current level of protection and durability across the approximately -10°C to 45°C outdoor-use range and assess cracking after relevant impacts or falls and deformation that could interfere with the brake mechanism; no numerical stiffness target exists."

Translated wording: "The need is supported by 17 prior-winter returns, including 13 from regions that had fallen below 0°C, with most cracks near the screw boss or wall-to-brake-support junction, and by an informal observation in which two of five housings cracked after complete rollers were conditioned for four hours at -10°C and dropped 1.0 metre onto concrete; actual field histories were unknown, and the observation lacked a room-temperature control, measured impact energy and a standardized protocol."

Shared verified source evidence

"The new component should provide at least the same level of protection and durability as the current version." Page 1
"The main problem to solve is cracking reported during winter use, particularly after impacts or falls." Page 1
"The expected outdoor-use range is approximately -10°C to 45°C." Page 1
"17 cracked components were returned during the previous winter season." Page 2
"13 of the 17 returns came from regions where temperatures had fallen below 0°C." Page 2
"The real impact conditions and product history are unknown." Page 2
"Most cracks appeared near the screw boss or at the junction between the main wall and the brake support." Page 2
"Five complete rollers were conditioned for four hours at -10°C." Page 2
"Each roller was dropped from 1.0 metre onto concrete in several informal orientations." Page 2
"Two of the five brake housings cracked." Page 2
"No room-temperature control group was tested." Page 2
"The exact impact energy was not measured and the protocol is not standardized." Page 2
"The component must not deform enough to interfere with the brake mechanism. No numerical stiffness target exists." Page 2
"Component mass 82 g Confirmed" Page 1
"A mass reduction of approximately 15% would be commercially valuable, but reliability has higher priority." Page 1
"At least 30% recycled content by mass is expected." Page 2
"The visible surface should remain black and low-gloss and should not look noticeably lower quality." Page 1

Human review

APPROVED AS GENERATED

No changes made. The model translation and shared evidence above are the reviewed result.

4. Material-relevant criteria

Model-generated translation

The industrial baseline is injection moulding of an impact-modified polypropylene of unknown exact grade, with 2.5 mm nominal walls, two screws and one snap-fit; the current mould and machine were developed for injection-moulded polypropylene, and mould reuse is strongly preferred. Screening criteria include protection and durability in the stated service context, resistance to brake-interfering deformation, behaviour around the screw boss where whitening is sometimes visible, shrinkage compatible with dimensions and assembly, flow through thin mould sections despite no validated minimum-flow requirement, and mould-wear implications because abrasive reinforcements may increase wear. Sustainability and perceived-quality criteria retain the expected at-least-30% recycled content, preferred post-consumer feedstock, mandatory origin and traceability documentation, and black, low-gloss, not noticeably lower-quality surface; industrial comparison should retain the confirmed 38-second cycle with increases above 10% to be avoided, estimated approximately €2.10/kg current material price, estimated approximately 3% scrap rate, forecast approximately 180,000-part annual volume, and unknown current density.

Uncertainties

- The exact current grade, current density, validated minimum flow requirement and acceptable shrinkage range are unavailable.

Model critic assessment (before human review)

Revise: Most industrial, sustainability, appearance, shrinkage, flow, wear, and process-baseline clauses are supported by this section's own citations. However, the section locally asserts that mould reuse is strongly preferred and includes protection, durability, and brake-interfering deformation as screening criteria without citing the corresponding brief statements in this section. Citations in other dossier sections do not cure this same-object evidence gap.

Semantic support: Partial

Issue types: Evidence support

Translated wording: "the current mould and machine were developed for injection-moulded polypropylene, and mould reuse is strongly preferred."

Translated wording: "Screening criteria include protection and durability in the stated service context, resistance to brake-interfering deformation"

Shared verified source evidence

"Manufacturing process Injection moulding Confirmed" Page 1
"Current material Impact-modified polypropylene; exact grade unknown Partially known" Page 1
"Nominal wall thickness 2.5 mm Confirmed" Page 1
"Assembly Two screws and one snap-fit Confirmed" Page 1
"The current mould and machine were developed for an injection-moulded polypropylene." Page 2
"Large changes in shrinkage may create dimensional or assembly issues. Abrasive reinforcements may increase mould wear." Page 2
"The mould has thin flow sections, but no validated minimum flow requirement is available." Page 2
"Visible whitening sometimes appears around the screw boss." Page 2

"Current cycle time 38 seconds Confirmed" Page 1
"A cycle-time increase above 10% should be avoided." Page 2
"Current material price Approximately €2.10/kg Estimate" Page 1
"Current scrap rate Approximately 3% Estimate" Page 1
"Annual volume Approximately 180,000 parts Forecast" Page 1
"Current material density Unknown Not available" Page 1
"At least 30% recycled content by mass is expected. Post-consumer recycled content is preferred. Feedstock origin and traceability must be documented." Page 2
"The visible surface should remain black and low-gloss and should not look noticeably lower quality." Page 1

Human review

CORRECTED

Human-reviewed translation

The industrial baseline is injection moulding of an impact-modified polypropylene of unknown exact grade, with 2.5 mm nominal walls, two screws and one snap-fit; the current mould and machine were developed for injection-moulded polypropylene, and mould reuse is strongly preferred. Screening criteria include protection and durability in the stated service context, resistance to brake-interfering deformation, behaviour around the screw boss where whitening is sometimes visible, shrinkage compatible with dimensions and assembly, flow through thin mould sections despite no validated minimum-flow requirement, and mould-wear implications because abrasive reinforcements may increase wear. Sustainability and perceived-quality criteria retain the expected at-least-30% recycled content, preferred post-consumer feedstock, mandatory origin and traceability documentation, and black, low-gloss, not noticeably lower-quality surface; industrial comparison should retain the confirmed 38-second cycle with increases above 10% to be avoided, estimated approximately €2.10/kg current material price, estimated approximately 3% scrap rate, forecast approximately 180,000-part annual volume, and unknown current density.

Reviewed uncertainties

- The exact current grade, current density, validated minimum flow requirement and acceptable shrinkage range are unavailable.

Reason for correction

Add same-section source support for mould reuse, comparative protection and durability, and brake-interfering deformation.

Evidence added or removed

Added evidence

"Existing mould Strong preference to reuse it Stakeholder preference" Page 1
"The new component should provide at least the same level of protection and durability as the current version." Page 1
"The component must not deform enough to interfere with the brake mechanism. No numerical stiffness target exists." Page 2

5. Missing information

Model critic assessment (before human review)

Revise: The section consolidates most important unknowns, including current-grade data, field history, validation methods, geometry, torque, flow, shrinkage, cost baseline, appearance tolerances, and documentation expectations. The phrase “any proposed material” broadens evidence about one supplier proposal to all possible materials. The section also does not explicitly identify the unresolved method or evidence needed to verify compliance with the required recycled-content fraction by mass.

Semantic support: Partial

Issue types: Evidence support, Omission

Translated wording: "Specify acceptable feedstock-origin and traceability documentation and obtain application evidence for any proposed material, including current-mould evidence absent from the supplier suggestion."

Human review: CORRECTED

ITEM 1 - APPROVED AS GENERATED

Establish the exact current-grade and density baselines, comparable material or finished-part evidence, actual field impact histories, and the relative contributions of material, local geometry and assembly stress.

Source evidence: page 1, page 2

ITEM 2 - APPROVED AS GENERATED

Define repeatable validation methods, controls and acceptance criteria for cracking, impact durability, brake-interfering deformation, “the same level of protection and durability,” and not feeling less robust; the existing cold-drop observation did not measure exact impact energy, was not standardized and had no room-temperature control, while no numerical stiffness target exists.

Source evidence: page 1, page 2

ITEM 3 - APPROVED AS GENERATED

Resolve the screw-boss geometry under review and obtain the assembly-torque, dimensional, shrinkage and minimum-flow requirements needed to judge compatibility with the component, assembly and existing mould.

Source evidence: page 1, page 2

ITEM 4 - APPROVED AS GENERATED

Define the component-cost baseline, scope and allowable increase beyond the approximately €2.10/kg estimated material-price proxy, along with acceptable manufacturing disruption, black and low-gloss tolerances, the meaning of “noticeably lower quality,” and what constitutes a significant durability decrease.

Source evidence: page 1, page 2

ITEM 5 - CORRECTED

Model-generated item

Specify acceptable feedstock-origin and traceability documentation and obtain application evidence for any proposed material, including current-mould evidence absent from the supplier suggestion.

Human-reviewed item

Specify acceptable feedstock-origin and traceability documentation and obtain application evidence for the supplier-proposed material, including current-mould evidence absent from the supplier suggestion.

Reason: Limit the application-evidence gap to the supplier proposal described by the cited source.

Source evidence: page 2

ITEM 6 - ADDED

Define the method and supporting evidence for verifying the required recycled-content fraction by mass.

Addition reason: Make the unresolved recycled-content verification method and supporting evidence explicit.

Source evidence: page 2

6. Conflicts and trade-offs

Model critic assessment (before human review)

Revise: The section captures the principal competing priorities and appropriately treats the supplier proposal as unsupported application advice. However, “cost ... should also be avoided” loses the source’s limitation to a significant increase in component cost and changes a team desire into a stronger normative formulation. “A material decision cannot be isolated” is also more absolute than the source’s qualified statement that the root cause may involve material, geometry, assembly stress, or a combination.

Semantic support: Partial

Issue types: Qualifier force

Translated wording: "The commercially valuable approximately 15% mass reduction is subordinate to reliability; recycled content is not considered successful if durability decreases significantly, while cost and manufacturing disruption should also be avoided."

Translated wording: "A material decision cannot be isolated from local geometry and assembly stress because the root cause may involve any combination of them, the screw-boss geometry may change, and screw torque is not systematically recorded."

Human review: CORRECTED

ITEM 1 - CORRECTED

Model-generated item

The commercially valuable approximately 15% mass reduction is subordinate to reliability; recycled content is not considered successful if durability decreases significantly, while cost and manufacturing disruption should also be avoided.

Human-reviewed item

The commercially valuable approximately 15% mass reduction is subordinate to reliability; recycled content is not considered successful if durability decreases significantly, while the team wants to avoid a significant increase in component cost or disruption to manufacturing.

Reason: Restore the source's significant-cost threshold and team-preference wording.

Source evidence: page 1, page 2

ITEM 2 - CORRECTED

Model-generated item

A material decision cannot be isolated from local geometry and assembly stress because the root cause may involve any combination of them, the screw-boss geometry may change, and screw torque is not systematically recorded.

Human-reviewed item

Material decisions should account for the unresolved possible contributions of material, local geometry and assembly stress because the root cause may involve any of them or a combination; the screw-boss geometry may change, and screw torque is not systematically recorded.

Reason: Replace an absolute causal statement with the source's unresolved, multi-factor framing.

Source evidence: page 1, page 2

ITEM 3 - APPROVED AS GENERATED

The supplier’s proposed 30% glass-fibre-reinforced PA6 and claim that it “should solve the issue” remain unsupported application suggestions: the supplier has not tested it in the current mould or supplied application data, while the mould and machine were developed for injection-moulded polypropylene, mould reuse is strongly preferred, shrinkage changes may cause dimensional or assembly issues, and abrasive reinforcement may increase mould wear.

Source evidence: page 1, page 2

Evidence appendix - list sections

Full source quotations are printed once here. Item cards use page references to avoid duplicating the same evidence.

Missing information item 1

"Current material
Impact-modified polypropylene; exact grade unknown
Partially known"

Page 1

Missing information item 1

"Current material density
Unknown
Not available"

Page 1

Missing information item 1

"No standardized material-specimen data is available for the current grade. Existing evidence comes from finished parts, field returns and internal observations."

Page 1

Missing information item 1; Conflicts and trade-offs item 2 (model); Conflicts and trade-offs item 2 (human)

"The root cause may involve the material, local geometry, assembly stress or a combination."

Page 2

Missing information item 1

"The real impact conditions and product history are unknown."

Page 2

Missing information item 2

"The exact impact energy was not measured and the protocol is not standardized."

Page 2

Missing information item 2

"No room-temperature control group was tested."

Page 2

Missing information item 2

"No numerical stiffness target exists."

Page 2

Missing information item 2

"The new component should provide at least the same level of protection and durability as the current version."

Page 1

Missing information item 2

"The component must not feel less robust than the current one."

Page 2

Missing information item 3; Conflicts and trade-offs item 2 (model); Conflicts and trade-offs item 2 (human)

"The local geometry around the screw boss is still being reviewed and may change."

Page 1

Missing information item 3; Conflicts and trade-offs item 2 (model); Conflicts and trade-offs item 2 (human)

"Screw torque is not systematically recorded."

Page 2

Missing information item 3

"Large changes in
shrinkage may create dimensional or assembly issues."

Page 2

Missing information item 3

"The mould has thin flow sections, but no validated minimum
flow requirement is available."

Page 2

Missing information item 4; Conflicts and trade-offs item 1 (model); Conflicts and trade-offs item 1 (human)

"The team wants to avoid a significant increase in component cost or disruption to manufacturing."

Page 1

Missing information item 4

"Current material price
Approximately €2.10/kg
Estimate"

Page 1

Missing information item 4

"The visible surface should remain black and low-gloss and should not look noticeably lower quality."

Page 1

Missing information item 4; Conflicts and trade-offs item 1 (model); Conflicts and trade-offs item 1 (human)

"Recycled content is not a success if durability decreases
significantly."

Page 2

Missing information item 5 (model); Missing information item 5 (human)

"Feedstock
origin and traceability must be documented."

Page 2

Missing information item 5 (model); Missing information item 5 (human)

"We
have not tested it in the current mould and have not provided supporting data for this application."

Page 2

Missing information item 6 (human)

"At least 30% recycled content by mass is expected. Post-consumer recycled content is preferred. Feedstock
origin and traceability must be documented."

Page 2

Conflicts and trade-offs item 1 (model); Conflicts and trade-offs item 1 (human)

"A mass reduction of approximately 15% would be commercially valuable, but reliability has higher priority."

Page 1

Conflicts and trade-offs item 3

"“We recommend a 30% glass-fibre-reinforced PA6 because it is stronger and should solve the issue. We
have not tested it in the current mould and have not provided supporting data for this application.”"

Page 2

Conflicts and trade-offs item 3

"The current mould and machine were developed for an injection-moulded polypropylene."

Page 2

Conflicts and trade-offs item 3

"Large changes in
shrinkage may create dimensional or assembly issues. Abrasive reinforcements may increase mould wear."

Page 2

Conflicts and trade-offs item 3

"Existing mould
Strong preference to reuse it
Stakeholder preference"

Page 1

Review conclusion

Final verdict: APPROVED WITH CORRECTIONS

Change summary

- Product intent: Corrected

- Component function: Approved as generated

- Performance to validate: Approved as generated

- Material-relevant criteria: Corrected

- Missing information: Corrected (Approved as generated: 4, Corrected: 1, Added: 1)

- Conflicts and trade-offs: Corrected (Corrected: 2, Approved as generated: 1)

Deterministic validation

Valid contents: 13; invalid contents: 0.

Known limitations

- Mechanical revalidation proves exact citation, source-reference and numeric provenance rules; it does not prove engineering correctness.

- The model critic is not rerun over human revisions; any semantic evaluation is a separate supporting artifact and is not part of this review record.

- Approval records one reviewer decision and is not independent multi-reviewer sign-off.

Artifact references and hashes

`Source SHA-256: 43b394906f3fc409e977a2e3462c65d8c778ec159489e5dd10ea3b15b81f7300`

`Base report JSON SHA-256: 3a00227fc8da9f5fa5495d86ea84eb56915e12dc6c6fb1025c0ae2d38dc92fb0`

`Reviewed JSON SHA-256: e0c508fdb9fbf5eff06970a5a9fe688a64b53e6f9d49f73ff94c13451cb37796`

The sibling manifest.json records the finalized Markdown and PDF SHA-256 values after those files are written. The PDF does not embed its own hash because that would change the PDF bytes.